

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Organization and Management

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. ☒ a) Define management, explain its functions and prove that management is universal in its application. 8
☒ b) Compare and contrast formal organization with informal organization with examples. 7
2. ☒ a) Describe the principles of organization. 8
☒ b) As a manager, how would you motivate your employees? Explain using a motivational theory. 7
3. ☒ a) Discuss the concept of participative management and its impact on employee motivation. 8
☒ b) Define human resource management and describe its key functions. 7
4. ☒ a) Why training is needed to develop human resource? Explain the methods of training. 8
☒ b) Define industrial relations and discuss the role of trade unions in collective bargaining. 7
5. ☒ a) Explain the major challenges of industrial relations in Nepal and suggest methods to improve them. 8
☒ b) As a management expert, how would you manage conflicts? Explain using conflict resolution techniques and approaches to conflict. 7
6. ☒ a) Discuss the potential negative impacts of poor conflict management on an organization's human resource management. 8
☒ b) Define collective bargaining and describe the process of it. 7
7. Write short notes on: (Any two) 2×5
 - ☒ a) Arbitration in conflict management
 - ☒ b) Job description and specification
 - ☒ c) Leader vs. Manager

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2024
Programme: BE Full Marks: 100
Course: Social and Professional Issues in IT Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Case Study:

Dark Web Malware Logs Expose 3,300 Users Linked to Child Abuse Sites

An analysis of information-stealing malware logs published on the dark web has led to the discovery of thousands of consumers of child sexual abuse material (CSAM), indicating how such information could be used to combat serious crimes.

"Approximately 3,300 unique users were found with accounts on known CSAM sources," Recorded Future said in a proof-of-concept (PoC) report published last week. "A notable 4.2% had credentials for multiple sources, suggesting a higher likelihood of criminal behavior."

Over the past few years, off-the-shelf info-stealer variants have become a pervasive and ubiquitous threat targeting various operating systems with an aim to siphon sensitive information such as credentials, cryptocurrency wallets, payment card data, and screenshots.

This is evidenced in the rise of new stealer malware strains such as Kematan Stealer, Neptune Stealer, Obj3ctivity, Poseidon (formerly RodStealer), Satanstealer, and StrelaStealer.

Distributed via phishing, spam campaigns, cracked software, fake update websites, SEO poisoning, and malvertising, data harvested using such programs typically find their way onto the dark web in the form of stealer logs from where they are purchased by other cybercriminals to further their schemes.

"Employees regularly save corporate credentials on personal devices or access personal resources on organizational devices, increasing the risk of infection," Flare noted in a report last July.

"A complex ecosystem exists in which malware-as-a-service (MaaS)

vendors sell info-stealer malware on illicit Telegram channels, threat actors distribute it through fake cracked software or phishing emails, and they then sell infected device logs on specialized dark web marketplaces."

Recorded Future's Insikt Group said it was able to identify 3,324 unique credentials used to access known CSAM domains between February 2021 and February 2024, using them to unmask three individuals who have been found to maintain accounts at no less than four websites.

The fact that stealer logs also comprise cryptocurrency wallet addresses means they could be used to determine if the addresses have been used to procure CSAM and other harmful material.

Furthermore, countries like Brazil, India, and the U.S. had the highest counts of users with credentials to known CSAM communities, although the company said that it could be due to an "overrepresentation due to dataset sourcing."

"Info-stealer malware and stolen credentials are projected to remain a cornerstone of the cybercriminal economy due to the high demand by threat actors seeking initial access to targets," it said, adding it has shared its findings with law enforcement.

"Info-stealer logs can be used by investigators and law enforcement partners to track child exploitation on the dark web and provide insight into a part of the dark web that is especially difficult to trace."

Answer the following questions relating the above case study:

- a) From the ethical, moral and legal perspective how do you relate the above examples? Discuss the involvement of IT personnel in such activities. 8
- b) How has above case and similar scenario affected the concept of privacy? Put your views. 7
2. a) What is digital divide? Discuss how covid pandemic enlarged the digital divide. 8
- b) What are professional rights and professional responsibilities? Explain violations of professional responsibilities with few examples. 7
3. a) How can we be safe while working in the internet? Discuss. 5
- b) What is digital forensic? Explain the process of digital forensics. 5
- c) How are copyright and patent violations dealt in Nepal? Explain in brief. 5
4. a) Write down the brief history of the Internet development. 7
- b) Deontological and Utilitarian based theories do not necessarily come 8

- to the same ethical decision when applied to an ethical argument. Justify with an example.
5. a) ✓ "Professional rights come along with responsibilities." Explain this in the context of Engineering. 7
- b) Analyze a complex software design problem and propose a solution that addresses both functional and non-functional requirements. 8
6. a) ✓ Discuss the strategies and measures that both governments and consumers can implement to protect sensitive information in the digital age. What is the role of government in protecting information? 7
- b) ✓ Examine the current threat landscape in cybersecurity and discuss the role of digital forensics in responding to cyber incidents. List out the types of digital forensic. 8
7. Write short notes on: (Any one) 10
- a) Define Intellectual Property (IP) rights and discuss the necessity of protecting IP. Additionally, analyze the global issues that arise in the context of IP.
- b) Define the concept of policy and discuss the process of formulating national policies. Additionally, provide an overview of five IT-related policies implemented in Nepal.
- c) ✓ How can IT professionals protect individual and organizational privacy? Discuss.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks

Semester: Spring

Year: 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain the merit and demerits of computer networks. List the merits of client-server model over peer to peer model. 7
- b) Define interfaces and services. Explain in brief OSI reference model. 8
2. a) What is Transmission media. How ISDN interfaces and channel works? 7
- b) Suppose you are a private consultant hired by a company to setup the network for their enterprise and you are given a large number of consecutive IP address starting at 120.89.96.0/19. Suppose that four departments A, B, C and D request 100, 500, 800 and 400 addresses respectively, how the subnetting can be performed so that address wastage will be minimum. 8
3. a) Explain Gigabit Ethernet with help of below parameters: Mac-sub layer, full duplex mode, half duplex mode, topology, implementation and encoding. 7
- b) Compare Stop-wait-ARQ, Go-Back-N and Selective Repeat based on below points 8
 - a. Mode of communication.
 - b. Addressing channel error
 - c. Handling error control and buffering
 - d. Technique for channel utilization
4. a) What is classful and classless address? Differentiate between link state and distance vector routing protocol. 7
- b) What are the functions of the Transport Layer? Explain TCP and its header segment. 8
5. a) What is congestion in a Network? Compare between leaky bucket and token bucket algorithm with the operation how token bucket works 7
- b) What is the role of URLs and Web Browsers in WWW. Explain about DHCP server in details. 8
6. a) What do you mean by Asymmetric key algorithm? Explain RSA algorithm Encryption and decryption process with suitable example 8

b) What is NMS. Explain about major components of Network Management Architecture. 7

7. Write short notes on: (Any two)

2×5

- a) HTTP
- b) IPv4 vs IPv6
- c) ATM

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Information System

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Information System. Discuss different information system components? What are different quality requirements that an information system should possess 8
b) What are different threats to Information System in terms of Security? Discuss different controls and security measures (including physical and logical) that are to be applied to project Information Systems? 7
2. a) Enterprise Management System is concerned with control, monitoring and the management of IT infrastructure and applications in order to optimize IT service delivery in Company. Give your opinions. 7
b) Who is enterprise engineer and what is enterprise engineering? How Customer Relationship Management form bonds between a company and its customers. 8
3. a) Describe Group Decision Support Systems (GDSS) for modern organizations. How do they help organization learning? 8
b) What is Data Warehouse? Why do you require data warehouse along with operational databases? How is it related with data mining? 7
4. a) How IS are used to deals with creation and implementation of tactics? Put your views. 7
b) What do you mean by Critical Success Factor (CSF) Explain various types of CSF with Examples. 8
5. a) What are the key dimensions of change management? Explain change management process. 7
b) What is collaborative filtering? How are we using such filters in our daily life? Explain. 8

6. a) Why is navigation important on a website? State the golden rules of web navigation. 7

b) Define Cloud Computing. Classify it based on access type and service type. 8

7. Write short notes on: (Any two) 2×5

a) Information Retrieval in cloud

b) Next Generation Balance Score Card

c) Loose Integration Vs Full Integration

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Signal Analysis and Processing

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the advantages of digital signal processing over analog signal processing? Briefly explain the basic elements of digital signal processing with the help of block diagram. 8
b) Obtain a linear convolution of the following two discrete-time signals:
 $x(n) = \sum_{k=0}^3 \delta(n-k)$ and $h(n) = 2^n[u(n)-u(n-3)]$ 7
2. a) Define the term causality, stability, time-invariance and linearity of discrete time system. And state their significance in case of LTI system. 7
b) Define Z transform. Find the inverse z transform of $X(z) = \log(1 + az^{-1})$. 8
3. a) Find the 8-point DIFFFT of $x(n) = \sin \frac{\pi}{3} n$. 8
b) Show that the multiplication of two DFT sequences results in circular convolution. 7
4. a) Find the direct form-I and direct form-II realizations of a discrete-time system represented by the transfer function
$$H(z) = \frac{3z^3 - 5z^2 + 9z - 3}{[z - (\frac{1}{2})][z^2 - z + (\frac{1}{3})]}$$
 8
b) How are FIR digital filters designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain. 7
5. a) Design a linear FIR filter using Kaiser window to meet the following specifications: 7
$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \text{ for } 0 \leq |\omega| \leq 0.19\pi$$
$$\leq |H(e^{j\omega})| \leq 0.01, \text{ for } 0.21\pi \leq |\omega| \leq \pi$$

b) Design a digital lowpass filter using Bilinear Transformation to satisfy the following requirements 8

- Monotonic stopband and passband
- -3dB cut off frequency at 0.6π radians, and
- Magnitude down at 15 dB at 0.75π radians

6. a) Convert the analog filter with system function

8

$$H_a(S) = \frac{S + 0.2}{(S + 0.2)^2 + 9}$$

into a digital filter by means of the bilinear transformation. The digital filter is to have a resonant frequency of $\omega_r = \pi/2$.

b) Differentiate between FIR and IIR digital filters with examples.

7

7. Write short notes on: (Any two)

2×5

- Linear convolution vs. circular convolution
- Causal System
- Remez Exchange Algorithm